

Sachindu Nanayakkara

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PROFESSIONAL SUMMARY

Final-year Information Technology undergraduate at the University of Moratuwa specializing in full-stack web development and scalable backend systems. Experienced in designing production-ready platforms featuring secure authentication, real-time communication, and high-performance APIs. Passionate about solving complex engineering problems and building reliable software using modern development practices.

CORE SKILLS

Languages: JavaScript, Python, Java, C

Frontend: React, Next.js (App Router), Tailwind CSS, Redux Toolkit

Backend: Node.js, Express.js, REST API Design, JWT Authentication, RBAC

Databases: MongoDB, MySQL, PostgreSQL, Redis

Realtime & Systems: Socket.IO, Event-Driven Architecture, Web Audio API

Cloud & tools: Cloudinary, SendGrid, Stripe, Git

Concepts: Scalable Architecture, Secure System Design, Performance Optimization

EXPERIENCE

ICIEOS

Feb 2025 – Jul 2025

Intern Software Engineer

Colombo, Sri Lanka

- Contributed to development of multiple full-stack web applications across music platforms, workflow systems, and digital distribution products
- Implemented frontend and backend features integrating APIs, real-time communication, and scalable workflows
- Collaborated in Agile teams to deliver production-ready features while maintaining performance and usability
- Participated in debugging, feature enhancements, and cross-team integration across ongoing client projects
- Tech Stack: React, Next.js, Node.js, Express.js, MongoDB, Redux Toolkit, Socket.IO, Redis, Stripe, Tailwind CSS

PROJECTS

Kumzits Everywhere | *Software Engineer (Intern)*

Feb 2025 – May 2025

- Live: Kumzits Everywhere
- Contributed to a large-scale music platform supporting bookings, events, e-commerce, and real-time messaging.
- Developed real-time chat interfaces with attachment support using React, Redux Toolkit, and Socket.IO.
- Implemented performance-optimized messaging using lazy loading and centralized state management.
- Engineered precision metronome using Web Audio API independent of JavaScript event loop timing.
- Contributed to scalable email microservice using Nodemailer, Redis queues, and SendGrid integration.
- Integrated Redis queues and SendGrid to support reliable bulk and transactional email delivery while reducing load on the primary application backend.
- Tech Stack: React, Node.js, MongoDB, Socket.IO, Redis, Nodemailer, SendGrid

Saubhagya Scholarship | *Full-stack Developer*

Jan 2026 – Current

- Designed and developed a role-based scholarship sponsorship platform enabling structured collaboration between students and donors.
- Implemented secure authentication and authorization using JWT access control and encrypted credential workflows.
- Built scalable backend APIs using Next.js Route Handlers and MongoDB models for users, donations and reports.
- Developed real-time communication system using Socket.IO with permission-based access rules.
- Implemented document management pipeline supporting validated PDF uploads via Cloudinary storage.
- Enabled Progressive Web App (PWA) functionality for installable mobile experience.
- Tech Stack: Next.js, React, Node.js, MongoDB, Tailwind CSS, Socket.IO, Cloudinary, JWT, Nodemailer

Tuition Management Platform | *Full-stack Developer*

Jan 2026 – Current

- Developed role-based classroom management system supporting academic content distribution and communication workflows.
- Built RESTful backend services with JWT authentication, RBAC authorization, and secure password recovery processes.
- Implemented moderated class forum system with configurable activation and soft-deletion logic.
- Designed class-restricted messaging infrastructure prepared for real-time Socket.IO communication.
- Developed responsive dashboards using Next.js App Router and Tailwind CSS.
- Tech Stack: Next.js, Node.js, Express.js, MongoDB, JWT, Socket.IO, Tailwind CSS

Supreme Edits DJ Pool & Music Distribution Platform | *Software Engineer (Intern)*

May 2025 – Jul 2025

- Live: Supreme Edits
- Contributed to development of scalable DJ pool platform enabling remix uploads, subscriptions, and content management.
- Implemented playlist APIs with advanced sorting and filtering to improve content discovery.
- Engineered track compatibility recommendation logic based on musical key and tempo attributes.
- Enhanced download cart system with Redux-Persist and MongoDB synchronization for cross-device continuity.
- Improved admin analytics by fixing tag usage tracking and reporting workflows.
- Tech Stack: React, Node.js, Express.js, MongoDB, Redux Toolkit, Redis

Green Haven | *Full-stack developer*

Aug 2024 – Dec 2024

- Developed a scalable e-commerce platform for a plant shop with dedicated customer and admin portals.
- Implemented product browsing, cart management, secure checkout, and real-time order tracking workflows.
- Built administrative dashboard for product management, order monitoring, and business analytics.
- Integrated secure online payments using Stripe payment processing.
- Designed microservices-based backend architecture with OAuth authentication and Redis session management.
- Tech Stack: Node.js, React, Docker, MSSQL, Redis, gRPC, RabbitMQ

EDUCATION

University of Moratuwa

BSc (Hons) in Information Technology

- CGPA: 3.42/4.0

Jun 2022 – Jun 2026

Moratuwa, Sri Lanka

Sivali Central College

Physical Science Stream

- GCE A/L (2020): ICT: A, Combined Mathematics: B, Physics: C

Jan 2012 – Oct 2020

Ratnapura, Sri Lanka

ACHIEVEMENTS

HackX - University of Kelaniya Finalist (Top 10)

SLIOT - University of Moratuwa Finalist (Top 10)

REFERENCES

Dr.Lochandaka Ranathunga

- lochandaka@uom.lk

Senior lecturer, Faculty of IT, UoM

Mr.Nipun Mullegamgoda

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Senior Software Engineer, Calcey Technologies